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Diversity of Rhinovirus and Enterovirus Infections Among Pediatric Patients Hospitalized in Wisconsin, 2022–2023

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ABSTRACT

Severe rhinovirus (RV)/enterovirus (EV) respiratory infections in pediatric patients increased in the US during 2022–2023. This study aimed to characterize RV/EV genetic diversity among hospitalized children in Wisconsin and describe demographic and clinical factors of patients by RV/EV type. During September 2022–April 2023, five Wisconsin pediatric hospitals submitted five RV/EV-positive specimens weekly to CDC. Viral typing was performed using partial viral protein (VP) 4/VP2 (RV) and VP1 (EV) sequencing. Wisconsin Department of Health Services abstracted demographic, comorbid condition, and hospitalization course information from patient medical records. Ninety-six patients were included in this investigation. RV-C ($n = 40$, 42%) and RV-A ($n = 31$, 32%) were the most frequently detected species, with 46 RV and 5 EV types identified. The median age was 2.5 (IQR: 0.7–5.5) years. Seventy-one (74%) children had at least one comorbid condition. Asthma was more common among children with RV-C infections ($n = 12$, 30%) compared with other RVs ($n = 3$, 9%) ($p = 0.02$). RV-A and RV-C were important contributors to severe pediatric respiratory infection in Wisconsin. Molecular typing showed virus divergence typically observed during season-long investigations. Year-round molecular characterization is needed to better understand the contribution of RV/EV types to severe pediatric respiratory illness.

1 | Introduction

Rhinoviruses (RVs) and enteroviruses (EVs) are genetically related and belong to the family *Picornaviridae* [1]. Commonly used clinical diagnostic tests (e.g., polymerase chain reaction (PCR assays) typically target shared conserved genomic regions among RV and EV, making RVs and EVs indistinguishable. RVs are the most common cause of viral respiratory tract infections in humans [1–4], usually resulting in mild self-limited illness, such as the common cold [1–8]. However, their contribution to more severe illness is being increasingly recognized with the wider use of PCR panels [2]. Poliovirus and non-poliovirus enteroviruses (NPEV) differ clinically, with polioviruses causing poliomyelitis [9] and NPEV causing a wider range of illnesses, including acute flaccid paralysis, myocarditis, conjunctivitis, and hand, foot, and mouth disease [1]. Only a minority of EVs cause respiratory infections, including EV-D68, which mainly circulates through respiratory routes, causing respiratory illness.

There is a large diversity within RVs and EVs. Seven RV and EV species circulate in the human population (RV-A, B, and C and EV-A, B, C, and D) [1, 10], with RV-A and RV-C being the most frequently detected [5]. Among RVs and EVs, more than 250 types exist. RV incidence tends to peak in the spring and fall months [1, 3, 5, 7] while EV incidence tends to peak in the summer in the United States [9].

In 2022, there was an increase in pediatric hospitalizations of patients with severe respiratory illness who tested positive for RV/EV in several regions in the United States [11], including in Wisconsin. To understand the epidemiology, diversity, and clinical characterization of RV/EV types causing more severe respiratory illness, the Wisconsin Department of Health Services (WI-DHS), the Centers for Disease Control and Prevention (CDC), and the Wisconsin State Laboratory of Hygiene (WSLH) performed medical chart reviews and molecular typing of RV/EV clinical specimens from a sample of pediatric patients admitted to the hospital with severe respiratory illness.

2 | Methods

2.1 | Subject Selection

Multipathogen respiratory PCR panels were performed at the discretion of the treating clinicians and included different combinations of RV/EV, adenovirus, seasonal coronaviruses (229E, HKU1, NL63, OC43), SARS-CoV-2, human metapneumovirus (HMPV), influenza A/B, parainfluenzas 1–4, respiratory syncytial virus (RSV), *Bordetella parapertussis*, *Bordetella pertussis*, *Chlamydia pneumoniae*, and *Mycoplasma pneumoniae*. The first five residual RV/EV-positive clinical specimens per week collected from hospitalized pediatric patients aged < 22 years during September 2022 through April 2023 were requested from five Wisconsin children's hospitals. Hospitals prioritized specimens from intensive care unit (ICU) patients. All residual nasopharyngeal (NP) swabs were sent to the CDC via the WSLH for further molecular characterization.

Specimens were excluded if collected during outpatient or procedural visits, > 2 weeks after hospital admission, or if no medical record was available. For individuals with multiple submissions during the study period, only the first specimen and corresponding medical record were used, and the excess specimens were excluded (Supplemental Figure).

2.2 | Laboratory Methods

Total nucleic acids were extracted from 200 µL of RV/EV-positive NP specimens using a NucliSens EasyMAG (BioMerieux) at CDC. Molecular typing was performed on the extracts by amplifying and sequencing the partial VP4/VP2 capsid gene region for RV [12] and the VP1 region for EV [13].

2.3 | Data Abstraction and Statistical Methods

Information abstracted from medical records included patient demographic characteristics, comorbid conditions, coinfections, course of illness, level of care (ICU or other), and highest level of respiratory support required (mechanical or nonmechanical ventilation). Mechanical ventilation was defined as the use of bilevel positive airway pressure (BiPap), continuous positive airway pressure (CPAP), an endotracheal (ET) tube, or tracheostomy. While we abstracted information on the diagnosis of bacterial infections or pneumonia, we did not abstract information related to antibiotic prescription or pulmonary imaging.

Comorbid conditions, level of care, course of illness, and coinfections were tabulated overall and by RV/EV species. Patient outcomes, including length of stay, level of care, and respiratory support required, were examined by RV/EV species and stratified by demographic characteristics. Chi-square or Fisher's exact tests were used to identify statistically significant differences between demographic groups, frequency of ICU admission, and respiratory support, as well as the relationship between RV species and asthma. Kruskal-Wallis tests were used to compare the length of stay by demographic groups and RV/EV species.

Statistical analyses were completed in SAS version 9.4. A graph showing the number of RV/EV specimens by type and week of specimen collection was created in RStudio (R and RStudio version 4.4.0). *P*-values less than 0.05 were considered statistically significant. All activities associated with this work were reviewed by CDC and determined to be public health surveillance, not human subjects research, and were conducted consistent with applicable federal law and CDC policy.[1] This project was, therefore, waived from IRB review.

3 | Results

Ninety-six patients/specimens were included in the analysis. Most patients (72%) were under 5 years of age (Table 1). Fifty-two (54%) individuals identified as White and 73 (76%) as not Hispanic or Latino. There were 71 (74%) patients with at least one comorbid condition, which varied significantly by age group ($p < 0.01$). A greater proportion of those ≥ 2 years old had at least one comorbidity (85%) compared to those < 2 years old (60%) ($p < 0.01$) (data not shown). Asthma (23%), premature birth (17%), and gastrointestinal conditions (16%) were the most common comorbid conditions (Table 3). Among those with RV-C, 30% ($n = 12$) had asthma compared to 9% ($n = 3$) of those with RV-A or B ($p = 0.02$).

The median time from hospital admission to specimen collection was < 1 day and 56% of individuals were admitted to the ICU upon admission to the hospital (Table 1). Ninety-three people had respiratory panel test results detailed in their chart (Supplemental Table 1). Fifteen (16%) people were also infected with a non-RV/EV respiratory virus, including one individual coinfected with four respiratory viruses in addition to RV/EV. The most common co-infecting virus was RSV ($n = 7$, 7%). Other notable coinfecting viruses included adenovirus (5%), seasonal coronaviruses (2%), parainfluenza (2%), influenza (1%), and SARS-CoV-2 (1%). There were no detected coinfections with bacteria known to cause respiratory illness, though chart review revealed 10 individuals (10.4%) with a diagnosed bacterial infection, of which 4 were described as secondary to a primary viral infection.

The median length of stay in the hospital was 2 days (IQR: 1–6 days) (Table 2). Fifty-nine (62%) people required respiratory support, with 31 (53%) of them receiving mechanical ventilation. The proportion of patients requiring ICU admission, respiratory support, mechanical ventilation, and length of stay did

TABLE 1 | Demographics of Patients by Rhinovirus/Enterovirus (RV/EV) Species.

	Total		Rhinovirus A		Rhinovirus B		Rhinovirus C		Enterovirus D68		Coxsackievirus		Multiple RV/EV		Not Typed	
	***n* = 96**	**%**	***n* = 31**	**%**	***n* = 4**	**%**	***n* = 40**	**%**	***n* = 2**	**%**	***n* = 6**	**%**	***n* = 3**	**%**	***n* = 10**	**%**
Sex																
Male	48	50.0	16	51.6	3	75.0	19	47.5	1	50.0	4	66.7	2	66.7	3	30.0
Female	48	50.0	15	48.4	1	25.0	21	52.5	1	50.0	2	33.3	1	33.3	7	70.0
Age, median (IQR)	2.5 (0.7–5.5)		2.3 (0.6–6.5)		3.7 (2.2–6.9)		2.1 (0.6–4.1)		1.7 (0.1–3.4)		0.6 (0.1–2.7)		8.0 (5.2–8.3)		4.1 (1.2–6.3)	
Age Group																
< 2 years	42	43.8	15	48.4	0	0.0	19	47.5	1	50.0	4	66.7	0	0.0	3	30.0
2–4 years	27	28.1	6	19.4	2	50.0	14	35.0	1	50.0	1	16.7	0	0.0	3	30.0
5–11 years	21	21.9	6	19.4	2	50.0	6	15.0	0	0.0	0	0.0	3	100.0	4	40.0
12+ years	6	6.3	4	12.9	0	0.0	1	2.5	0	0.0	1	16.7	0	0.0	0	0.0
Race																
American Indian	3	3.1	1	3.2	0	0.0	1	2.5	1	50.0	0	0.0	0	0.0	0	0.0
Asian or Pacific Islander	4	4.2	2	6.5	1	25.0	1	2.5	0	0.0	0	0.0	0	0.0	0	0.0
Black or African American	22	22.9	7	22.6	0	0.0	13	32.5	0	0.0	1	16.7	1	33.3	0	0.0
White	52	54.2	18	58.1	3	75.0	18	45.0	0	0.0	4	66.7	2	66.7	7	70.0
Other	4	4.2	1	3.2	0	0.0	2	5.0	0	0.0	1	16.7	0	0.0	0	0.0
Missing/Unknown	11	11.5	2	6.5	0	0.0	5	12.5	1	50.0	0	0.0	0	0.0	3	30.0
Ethnicity																
Hispanic or Latino	12	12.5	4	12.9	0	0.0	6	15.0	0	0.0	0	0.0	0	0.0	2	20.0
Not Hispanic or Latino	73	76.0	25	80.6	4	100.0	29	72.5	1	50.0	6	100.0	3	100.0	5	50.0
Missing/Unknown	11	11.5	2	6.5	0	0.0	5	12.5	1	50.0	0	0.0	0	0.0	3	30.0
Comorbid Condition																
Any	71	74.0	19	61.3	3	75.0	32	80.0	2	100.0	4	66.7	3	100.0	8	80.0
Asthma	22	22.9	3	9.7	0	0.0	12	30.0	0	0.0	2	33.3	2	66.7	3	30.0
Admission Location																
ICU	54	56.3	19	61.3	1	25.0	22	55.0	2	100.0	3	50.0	1	33.3	6	60.0
Other	42	43.8	12	38.7	3	75.0	18	45.0	0	0.0	3	50.0	2	66.7	4	40.0
Time From Admission To Sample Collection, median (IQR)	0 (0–0)		0 (0–0)		0 (0–2.0)		0 (0–0)		0 (0–0)		0.5 (0–5.0)		0 (0–4.0)		0 (0–0)	
Coinfection*	15	15.6	3	9.7	2	50.0	4	10.0	0	0.0	1	16.7	1	33.3	4	40.0

Abbreviations: EV, Enterovirus; RV, Rhinovirus.
*Coinfection with non-RV/EV respiratory pathogen is tested for on the multipathogen respiratory PCR assay.

TABLE 2 | Outcomes by Rhinovirus/Enterovirus Species.

	Total		Rhinovirus A		Rhinovirus B		Rhinovirus C		Enterovirus D68		Coxsackievirus		Multiple RV/EV		Not Typed	
	n **= 96**	%	***n*** **= 31**	%	***n*** **= 4**	%	***n*** **= 40**	%	***n*** **= 2**	%	***n*** **= 6**	%	***n*** **= 3**	%	***n*** **= 10**	%
Length of Stay, median (IQR)	2.0 (1.0–6.0)		2.0 (1.0–8.0)		5.0 (2.5–6.0)		2.0 (1.0–3.5)		14.5 (1.0–28.0)		5.0 (1.0–14.0)		2.0 (0.0–59.0)		2.5 (0.0–6.0)	
Highest Level of Respiratory Support Required***																
Any	59	61.5	19	61.3	1	25.0	26	65.0	2	100.0	3	50.0	1	33.3	7	70.0
Mechanical*	31	32.3	13	41.9	1	25.0	10	25.0	1	50.0	2	33.3	0	0.0	4	40.0
Nonmechanical**	27	28.1	6	19.4	0	0.0	15	37.5	1	50.0	1	16.7	1	33.3	3	30.0

Abbreviations: EV, Enterovirus; RV, Rhinovirus.
*Mechanical ventilation included bilevel positive airway pressure (BiPAP), continuous positive airway pressure (CPAP), endotracheal tube (ETT), or tracheostomy needed at any point during the hospitalization.
**Nonmechanical respiratory support included oxygen mask, non-rebreather mask, or nasal cannula, including high flow nasal cannula, needed at any point during the hospitalization.
***One individual with a Rhinovirus C infection was missing information on the respiratory support required during hospitalization.

not differ significantly by age group or coinfection status (Table 3). Individuals who had been born prematurely ($n = 16$, 17%) were more likely to require respiratory support ($n = 14$, 24%, $p = 0.02$) and had longer lengths of stay (5.5 days, IQR: 2.5–22.5, $p = 0.01$) compared to those who had not been born prematurely ($n = 2$, 5% and 2.0 days, IQR 1.0–5.0, respectively). When the study population was restricted to children < 2 years old, those who had been born prematurely were more likely to require mechanical ventilation ($n = 7$ of 10, 70%) compared to those not born prematurely ($n = 8$ of 32, 25%) ($p = 0.02$) and had a longer length of stay (6.5 days, IQR: 3.0–28.0) compared with those not born prematurely (1.0 days, IQR: 2.0–7.5) ($p = 0.02$) (data not shown). The proportion of those needing respiratory support was also higher among those with premature birth ($n = 10$ of 10, 100%), compared with those without ($n = 16$ of 32, 50%) in this under 2 years of age group ($p < 0.01$).

RV was typed in 75 (78%) individuals, EV was typed in 8 (8%), and multiple RV/EV types in 3 (3%). Ten (10%) specimens could not be typed due to low viral titer. RV-C ($n = 40$, 42%) and RV-A ($n = 31$, 32%) were the most common species and were detected throughout the study period, while RV-B ($n = 4$, 4%) was rare. A total of 51 distinct types were identified (Figure 1), including 46 RV types in 78 patients and 5 EV types in 8 patients. The most common RV types were RV-C24 ($n = 5$) and RV-A21 ($n = 4$). Coxsackieviruses ($n = 6$, 6%), including CV-A16 and CV-A9 ($n = 2$ each) were detected in October and November. EV-D68 ($n = 2$, 2%) was identified only in September.

4 | Discussion

For our investigation into the epidemiology, diversity, and clinical characterization of RV and EV types in children in Wisconsin during the 2022–2023 season, we conducted enhanced surveillance of hospitalized pediatric patients with respiratory infections, focusing especially on those admitted to the ICU. Many circulating RV types were observed among children with severe respiratory illness, particularly among younger children and those with asthma. Specimens were collected from the most severe pediatric cases, but differences in clinical severity among certain groups of patients were seen.

Most individuals (84%) were only infected with RV or EV; few had coinfections. Consistent with the literature, RSV [2, 5, 6] and adenovirus [5, 6] were the most common coinfections with RV, and influenza was rarely detected with RV infection [2, 5, 6]. There were no detected coinfections with bacteria known to cause respiratory illness by the respiratory panel test, and only 10% of individuals had a diagnosed bacterial infection in their chart, four of which were described as secondary to a primary viral infection. While different pathogens were contained on different respiratory panels, more than 80% of people included were tested with a respiratory pathogen panel containing adenovirus, HMPV, influenzas A and B, parainfluenza viruses 1–4, RSV, seasonal coronaviruses, *Bordetella pertussis*, *Bordetella parapertussis*, *Chlamydia pneumoniae*, and *Mycoplasma pneumoniae*. With less than 20% of individuals in our study having a coinfection, admission to the hospital with respiratory symptoms for most patients was likely due to the RV/EV infection, rather than a different virus, suggesting that RV/EV infections can lead to severe illness in children. Most children in our investigation had at least one comorbid condition, but this did

TABLE 3 | Outcomes by Demographics.

	Total		Ever Admitted to ICU			Any Respiratory Support			Mechanical Ventilation*			Length of Stay		
	$n = 96$	**Column %**	**$n = 56$**	**Column %**	**p-value**	**$n = 59$**	**Column %**	**p-value**	**$n = 31$**	**Column %**	**p-value**	**Median**	**IQR**	**p-value**
Overall (row percents)	96	100%	56	58%	-	59	61%	-	31	32%	-	2.0	1.0–6.0	-
Sex					0.21			0.83			0.83			0.82
Male	48	50.0	25	44.6		30	50.8		16	51.6		2.0	1.0–5.0	
Female	48	50.0	31	55.4		29	49.2		15	48.4		2.0	1.0–7.0	
Age Group					0.07			0.88			0.09			0.33
< 2 years	42	43.8	24	42.9		26	44.1		15	48.4		3.0	1.0–10.0	
2–4 years	27	28.1	12	21.4		15	25.4		4	12.9		1.0	1.0–4.0	
5–11 years	21	21.9	14	25.0		14	23.7		9	29.0		2.0	1.0–5.0	
12+ years	6	6.3	6	10.7		4	6.8		3	9.7		2.5	2.0–5.0	
Coinfection					0.12			0.48			1.00			0.52
Yes	15	15.6	6	10.7		8	13.6		5	16.1		3.0	1.0–7.0	
No	81	84.4	50	89.3		51	86.4		26	83.9		2.0	1.0–5.0	
Comorbidities														
Any	71	74.0	41	73.2	0.84	47	79.7	0.11	24	77.4	0.59	3.0	1.0–7.0	0.05
Asthma	22	22.9	11	19.6	0.37	15	25.4	0.46	4	12.9	0.11	2.0	1.0–3.0	0.69
Gastrointestinal	15	15.6	8	14.3	0.67	8	13.6	0.48	4	12.9	0.77	3.0	2.0–8.0	0.23
Premature Birth	16	16.7	11	19.6	0.35	14	23.7	0.02	8	25.8	0.10	5.5	2.5–22.5	0.01

Abbreviation: IQR, Interquartile range.
*Mechanical ventilation included bilevel positive airway pressure (BiPAP), countinuous positive airway pressure (CPAP), endotracheal tube (ETT), or tracheostomy needed at any point during the hospitalization.

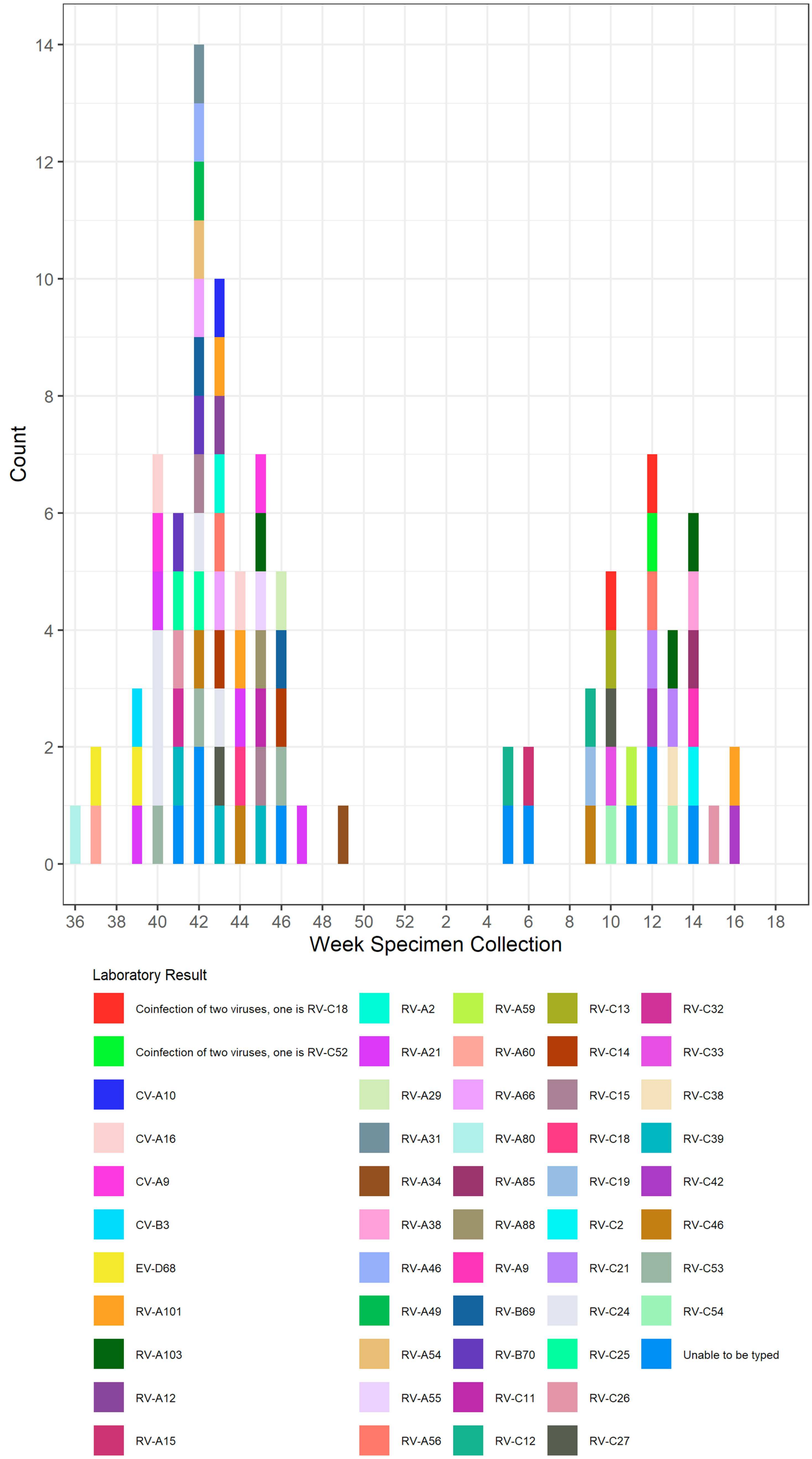

FIGURE 1 | Number of Rhinovirus/Enterovirus (RV/EV) Specimens by RV/EV Type and Week of Specimen Collection. RV = Rhinovirus, EV = Enterovirus, CV = Coxsackievirus.

not result in higher ventilation requirements or longer length of stays in the hospital. Individuals who had been born prematurely were especially impacted by RV/EV infection, with significantly higher requirements for respiratory support and longer lengths of stay in the hospital than those not born prematurely.

There were more specimens collected in the fall and spring than in the winter. These peaks more closely reflect RV seasonality as it is more common than EV, which tends to peak in the summer in the United States, during which we were not sampling for this investigation [9]. Most commercially available molecular tests do not distinguish between RV and EV, making it difficult to use testing data to determine seasonality. Viral typing with these specimens allowed us to look at the seasonality of different RV/EV species. Similar to other studies [1, 3, 5, 6], RV-C and RV-A were the predominant RV species observed in our study, affecting 40 (42%) and 31 (32%) patients, respectively. Both RV-A and RV-C circulated throughout the study period, with gaps during the holiday period when few specimens were received, and with peaks in the fall and spring, as seen in the literature [1, 3, 5], while RV-B, coxsackievirus, and EV-D68 were only seen in the fall season. Additionally, we found large type diversity, with 51 different RV/EV types circulating during the study period. There were no predominant types. This is consistent with previous studies that also showed high diversity with a lack of predominant type [3, 4, 10], but contrasts to certain years when outbreaks of EV-D68 occurred, such as 2014 [1, 9, 10].

Many of the individuals in our study had comorbidities, with asthma being the most common (23%). This was particularly prevalent in the population infected with RV-C where 30% of individuals had asthma compared to 10% of individuals with RV-A and none of the 4 individuals with RV-B ($p = 0.02$). Previous studies have investigated the connection between RV-C and asthma, with many finding that RV-C infection is associated with more severe asthma exacerbation and a higher hospitalization rate than RV-A and RV-B [4, 5, 10]. We did not see an association between asthma or other comorbidity status and low viral load that would suggest different care-seeking behavior for those with comorbidities, though our small sample size could have masked this association.

Our study had a few limitations. First, hospitals were asked to prioritize specimens from people in the ICU; however, the number and proportion of specimens from ICU patients varied by hospital. This left us unable to identify predictors of ICU admission (Supplemental Table 2). Additionally, the demographics of each hospital differed, potentially introducing confounding into our study results. For this reason, we were unable to examine patient demographics that varied by hospital and clinical severity. Second, because we asked for specimens from ICU patients to gain a better understanding of type diversity in severe RV/EV, we cannot comment on the proportion of severe cases as all specimens came from the most severe RV/EV cases. Finally, by sampling during September–April, we missed the circulation of some enteroviruses, including EV-D68. This resulted in too few individuals infected with EV-D68 to meaningfully comment on the characteristics and severity of disease caused by EV-D68.

Through typing of residual RV/EV-positive clinical specimens collected from hospitalized children, we showed the large diversity of circulating RVs and EVs, with 51 different RV and EV types detected. Largely, pediatric patients presented without other viral or bacterial coinfections, demonstrating the important contribution of RV and EV to clinically severe respiratory disease among children, particularly young children and those with asthma. A better understanding of RV and EV circulation and clinical outcomes would benefit from year-round surveillance to capture circulating enteroviruses, including EV-D68.

Author Contributions

Hannah L Litwak: formal analysis, writing – original draft, writing – review and editing. **Carlos M Perez:** data curation, data abstraction, formal analysis, writing – original draft. **Allen Bateman:** conceptualization/study design, data curation, writing – review and editing. **Kyley Anhalt:** conceptualization/study design, data curation, writing – review and editing. **Xiaoyan Lu:** data curation, writing – review and editing. **Shannon Rogers:** investigation, data curation, writing – review and editing. **Brian Emery:** investigation, data curation, writing – review and editing. **Terry Fei Fan Ng:** investigation, data curation, writing – review and editing, supervision. **Caelin Cubenas:** data analysis, writing – review and editing. **Cria O Gregory:** conceptualization/study design, writing – review and editing, supervision. **Clinton Paden:** data curation, writing – review and editing. **Thomas Haupt:** conceptualization/study design, data curation, writing – review and editing. **Hannah L Kirking:** conceptualization/study design, writing – review and editing. **Hannah E Segaloff:** conceptualization/study design, data curation, writing – review and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author as deemed appropriate by the Wisconsin Department of Health Services. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

[1] See e.g., 45 C.F.R. part 46.102(l)(2), 21 C.F.R. part 56; 42 U.S.C. §241(d); 5 U.S.C. §552a; 44 U.S.C. §3501 et seq

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.
Supplemental Figure: Subject Selection. **Supplemental Table I:** Coinfections by Rhinovirus/Enterovirus (RV/EV) Species. **Supplemental Table II:** Rhinovirus/Enterovirus (RV/EV) Species, Patient Demographics, and Outcomes by Hospital.